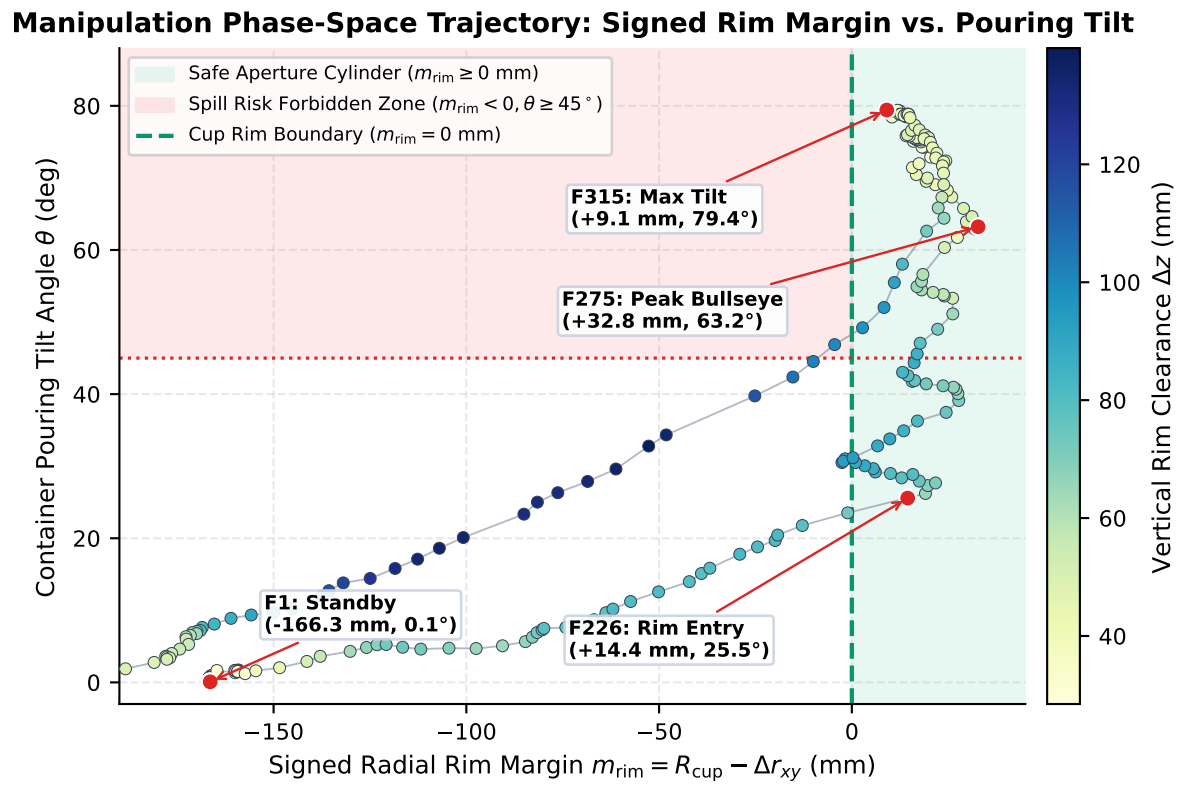

Manipulation Phase-Space Trajectory: Signed Rim Margin vs. Pouring Tilt
Safe Aperture Cylinder ($m_{rim} \geq 0$ mm)
Spill Risk Forbidden Zone ($m_{rim} < 0, \theta \geq 45^\circ$)
Cup Rim Boundary ($m_{rim} = 0$ mm)
F315: Max Tilt
(+9.1 mm, 79.4°)
F275: Peak Bullseye
(+32.8 mm, 63.2°)
F1: Standby
(-166.3 mm, 0.1°)
F226: Rim Entry
(+14.4 mm, 25.5°)
Container Pouring Tilt Angle θ (deg)
0
20
40
60
80
−150
−100
−50
0
Signed Radial Rim Margin $m_{rim} = R_{cup} - \Delta r_{xy}$ (mm)
Vertical Rim Clearance Δz (mm)
40
60
80
100
120